

ECOCIDE — Executive Summary

Flood, irrigation loss or war? A pre-registered, exposure-based satellite test of the vegetation effects of the Kakhovka Dam destruction

Sakshi D. Maske · Independent Geospatial Researcher · Code and data: github.com/sakshimaske303-commits/ECOCIDE · Full paper: [EC0_Research_Paper_v2.md](#)

Question

The Kakhovka Dam on the lower Dnipro was destroyed on 6 June 2023. Its destruction could have affected vegetation through three separate pathways:

- the flood downstream;
- the draining of the reservoir;
- the loss of irrigation water in the canal system the reservoir fed.

All of this happened inside an active war zone. The study asks which of these pathways left a measurable mark on vegetation, once the war around them is taken into account.

Why a second study

Study 1 compared the monthly NDVI of all of Kherson Oblast with four Romanian counties. It found a significant relative decline after June 2023 (-0.108 , $p = 0.037$), but could not attribute it to the dam:

- the flood covered only 2% of the oblast;
- pre-event trends already diverged;
- in a randomization test Kherson ranked second of five units ($p = 0.40$).

Design (Study 2)

- **Data.** MODIS NDVI at 250 m, 2016–2024 (2010–2015 used for the revision), analysed pixel by pixel.
- **Treatment defined by exposure.** Flooded land (UNOSAT), the former reservoir bed, and cropland near the reservoir's canal network (OpenStreetMap).
- **Comparisons inside the war zone.** Matched unflooded land on the same river bank; irrigated vs rainfed cropland, inside vs outside the canal zone.
- **Fixed in advance.** Every rule, including the words used to report a verdict, was set in `ANALYSIS_PLAN_v2.md`, committed before any data were downloaded (commit 521672f).
- **Registered revision.** After the first results exposed five design problems, a revision (`ANALYSIS_PLAN_v2_ADDENDUM_1.md`, commit 30e3840) was committed before its extra data were obtained. It:
 - measured irrigation from 2010–2015;
 - restricted the comparison zone to the steppe oblasts;
 - used district-level placebos;
 - added VIINA occupation and conflict intensity;
 - added annual water masks.
- **Inference.** Clustered and spatial (Conley) standard errors, wild cluster bootstrap, randomization tests against placebo units, Holm correction, and Rambachan–Roth sensitivity bounds.

Findings

Pathway	Pre-registered result	Registered revision	Verdict
Flood (H1): flooded vs matched unflooded land	+0.006 NDVI (CI -0.014 to 0.026); randomization $p =$	+0.055 (flooded land greener); $p = 0.88$ for a	Not supported

	0.63 across 141 placebo floodplains	decline	
Floodplain wetlands within the flood	−0.041	−0.061	Decline persists
Irrigation (H2): irrigated vs rainfed, canal zone vs elsewhere	−0.074, but strong pre-war divergence; randomization floor $p = 0.14$	−0.053, but no change after the breach relative to 2021 (−0.006 to −0.014, n.s.); 15 placebo districts, $p = 0.38$	Suggestive by rule; the divergence predates the invasion
Reservoir bed (H3)	Area with NDVI > 0.3: ~95 km ² (2016–22) → 906 km ² (2023) → 1,574 km ² (2024)	Same trajectory on land-only pixels	Clearest effect (descriptive)
Kherson Oblast change (H4), 2021→2024 vs comparison cropland	Total −0.029: other land −0.024, irrigated canal zone −0.013, flood −0.0004, reservoir bed +0.010	−0.052 with the water rule (reservoir bed excluded by construction)	Most of the decline is outside every dam pathway

What this means

1. **The flood itself did not leave a lasting loss of summer vegetation on the land it covered.** The exception is floodplain wetland, where greenness fell. That fall may reflect the lower, unregulated river after the dam's loss rather than the flood itself.
2. **The irrigation story is not supported once pre-war trends are handled.** The canal-zone decline happened between 2020 and 2021 and did not deepen after the breach. The NDVI-based irrigation measure also mostly captures summer cropping; an independent irrigation map would be needed.
3. **The drained reservoir became about 1,500 km² of new vegetation within two seasons.** Whether this is recovery or a new ecological and contamination risk is a question for field work.
4. **Study 1's oblast-wide decline was real, but mostly not caused by the dam.** An administrative-unit average mixes pathways with opposite signs; attribution needs units that match each pathway's footprint.

Limitations

- **Exposure measures.**
 - The flood polygons are UNOSAT's preliminary product.
 - Irrigation status is inferred from NDVI.
 - Canals come from current OpenStreetMap, not a dated pre-war map.
- **Common support (H1).** Few unflooded pixels resemble the flooded floodplain, so the matched H1 sample is small.
- **Pre-trends (H2).** Strong pre-war divergence in H2.
- **Annual water masks.** Coarse for 2023 and missing for 2024.
- **Conflict data.** VIINA is news-based.
- **Outcome.** NDVI measures greenness only.
- **Timing of the revision.** It was registered after the first results were known, so it is reported next to them, not instead of them.

Reproducibility

- Data download: v2/01-08.
- Pre-registered analysis: `python v2/analysis/run_all.py`.
- Revision: `python v2/analysis/run_revision.py`.
- Results: `outputs/v2/study2_summary.json` and `outputs/v2/r1_summary.json`.
- Every departure from the plan: v2/DEVIATIONS.md.